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Computer Network Protocols

Chapter Three

ICMP, DHCP, IPSec & Forwarding Methods

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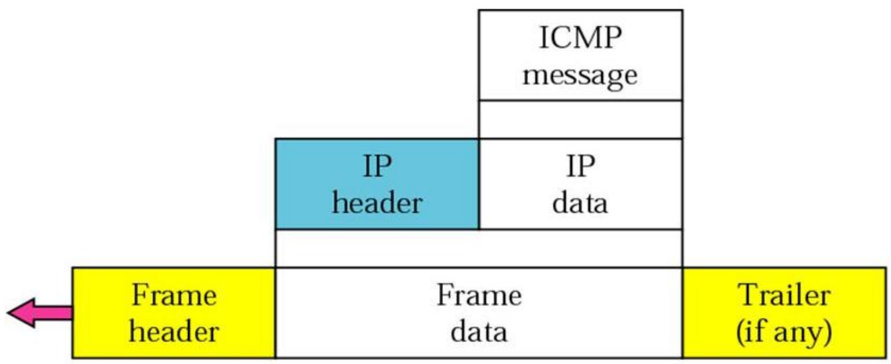
3.1- Internet Control Message Protocol (ICMP)

ICMP has been designed to compensate two deficiencies in IP protocol (Best effort delivery)

- ✓ Lack of error control
- ✓ Lack of assistance mechanism

3.1.1- ICMP Messages

ICMP itself is a network layer protocol. However, its messages are not passed directly to the data link layer as would be expected. Instead, the messages are first encapsulated inside IP datagrams before going to the lower layer.



ICMP messages are divided into two broad categories:

- **Error reporting messages:** report a problem that a router or a host (destination) may encounter during its processing to the IP packet.

- **Query messages:** help the host (source) or the network manager get specific information from a router or another host

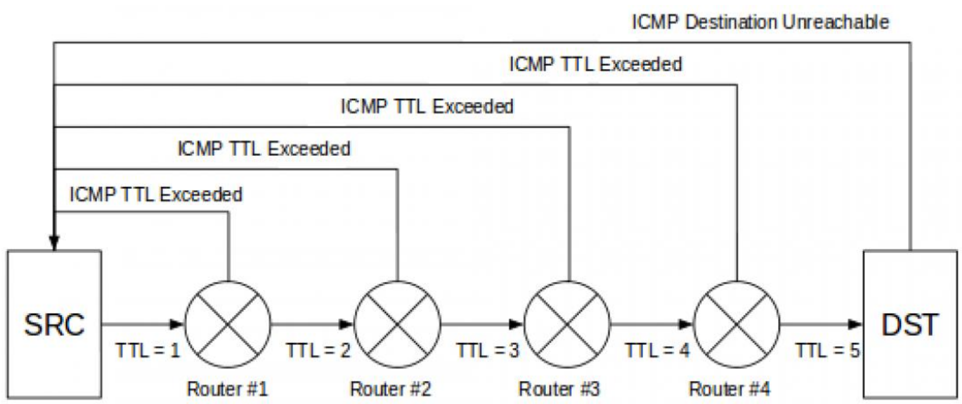
ICMP used by hosts & routers to communicate network-level information ICMP reports errors (unreachable host, router, port, or requested service is not available) and sends control message (echo request/reply (used by ping), ICMP does not attempt to make IP a reliable protocol. it simply attempts to report errors and provide feedback on the specific condition.

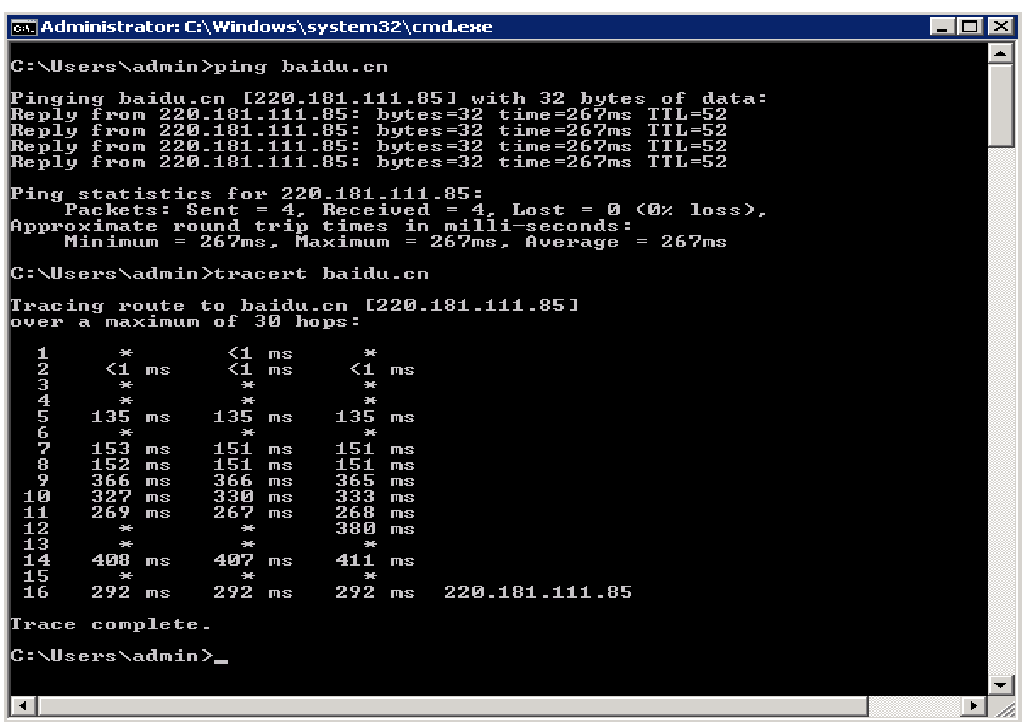
3.1.2- ICMP Applications

There are two simple and widely used applications that are based on ICMP:

Ping: The ping checks whether a host is alive & reachable or not. This is done by sending an ICMP Echo Request packet to the host and waiting for an ICMP Echo Reply from the host.

Traceroute: Traceroute is used to records the route through the Internet between your computer and a specified destination computer. It also calculates and displays the amount of time each hop took.



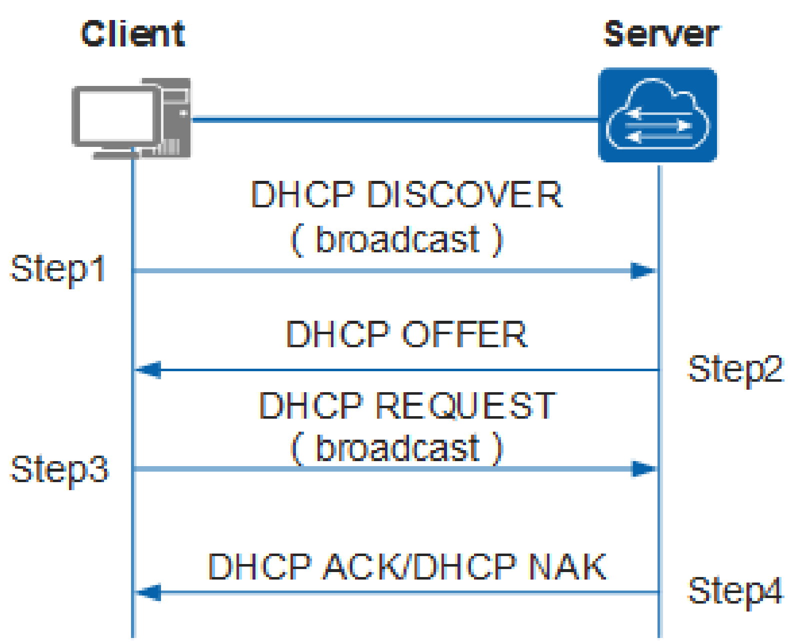


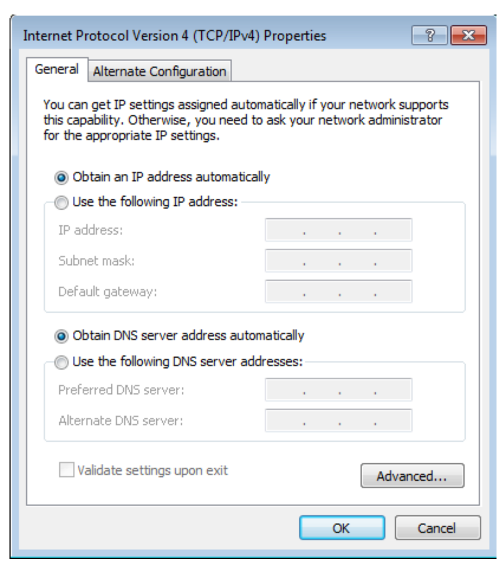
3.2- DHCP: Dynamic Host Configuration Protocol

Goal: allow a host to dynamically obtain its IP address from the network server when it joins the network.

DHCP overview: (pool operation)

- host broadcasts "**DHCP discover**" msg
- DHCP server responds with "**DHCP offer**" msg
- host requests IP address: "**DHCP request**" msg
- DHCP server sends address: "**DHCP ack**" msg



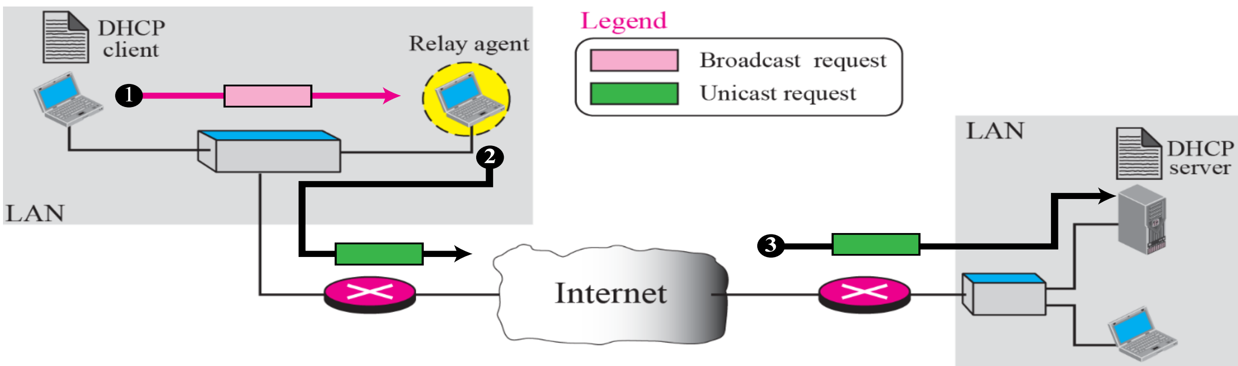


DHCP: more than IP addresses

DHCP can return more than just an allocated IP address on the subnet:

- ✓ address of the first-hop router for client
- ✓ name and IP address of DNS server
- ✓ network mask

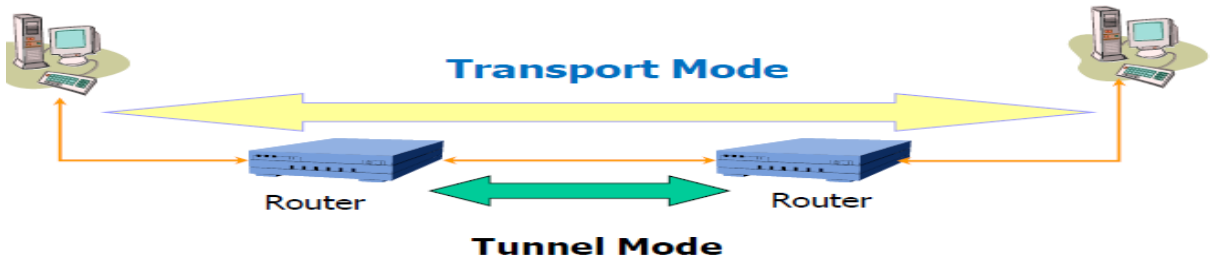
There is one problem in this method that must be solved. The DHCP request is broadcast because the client does not know the IP address of the server. A broadcast IP datagram cannot pass through any router.
To solve the problem, there is a need for an intermediary. One of the hosts (or a router) can be used as a relay. The host in this case is called a relay agent.

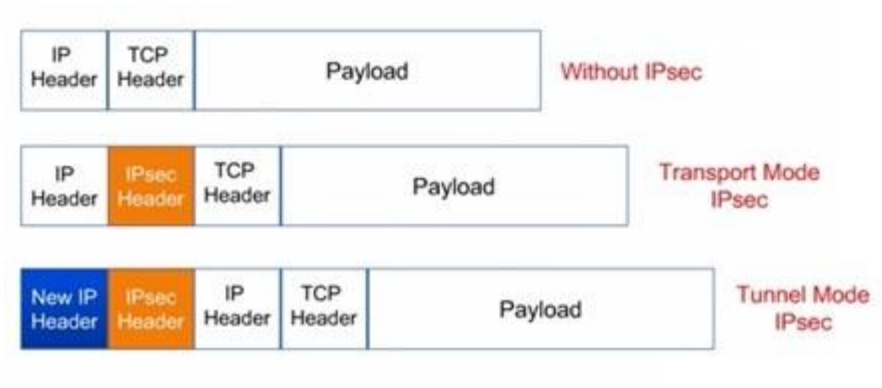


3.3- IP Security (IPSec)

- IP Security (IPSec) is a collection of protocols designed by the Internet Engineering Task Force (IETF) to provide security for a packet at the network level.
- IPSec helps to create authenticated and confidential packets for the IP layer.
- IPSec operates in one of two different modes: the **transport mode** or the **tunnel mode**.

Transport Mode	Tunnel Mode
IPSec in the transport mode does not protect the IP header; it only protects the information coming from the transport layer.	IPSec in the tunnel mode protects the original IP header.
IP Header not Encrypted	IP Header Encrypted
Used for end-to-end communications (client and server)	Most commonly used between routers or ASA firewall (gateways)

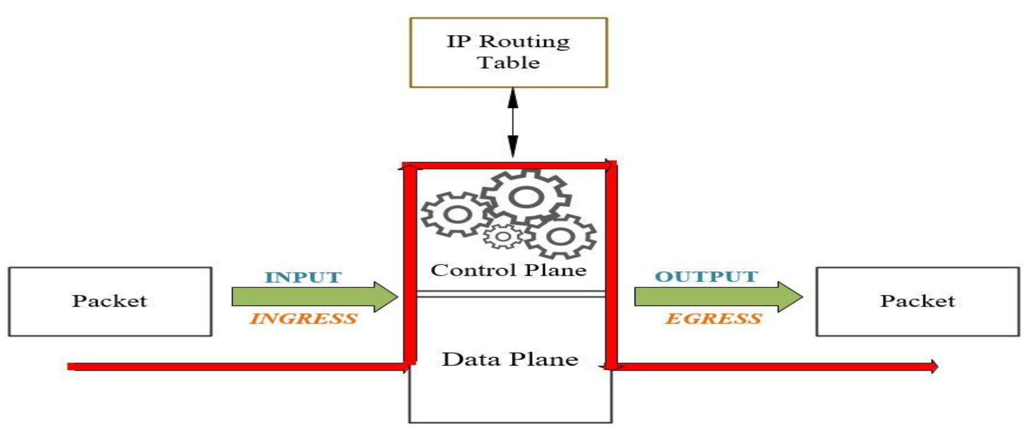




IPsec Packet format transport mode vs. tunnel mod

3.4- Packet Forwarding

Packet forwarding is the process by which a network device, typically a router, receives a data packet on an input interface and transmits it to the appropriate output interface based on the packet's destination address. It is the local operation of moving a packet within a single node, distinct from routing, which involves building the global tables used for these forwarding decisions.



Core Forwarding Mechanism

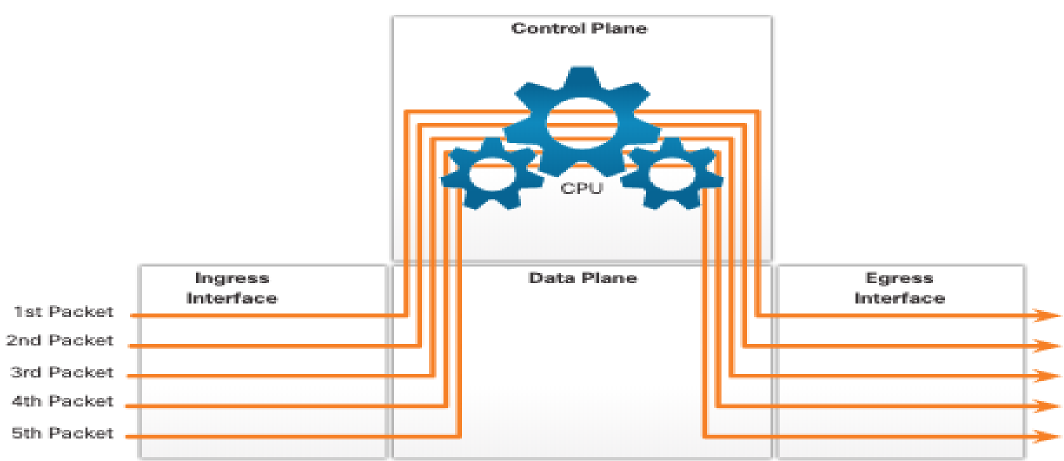
When an IP packet arrives at a router, it undergoes a series of standardized processing steps:

- **Validation**: The router verifies the IP header for correctness (e.g., checksum, version).
- **TTL Adjustment**: The Time-to-Live (TTL) field is decremented. If it reaches zero, the packet is discarded to prevent infinite loops.
- **Table Lookup**: The router extracts the destination IP address and performs a Longest Prefix Match (LPM) lookup in its forwarding table.
- **Next-Hop Determination**: The lookup identifies the next-hop neighbor or the exit interface.
- **Encapsulation & Transmission**: The packet is re-encapsulated with a new Layer 2 (MAC) header for the next link and placed in an output queue for transmission.

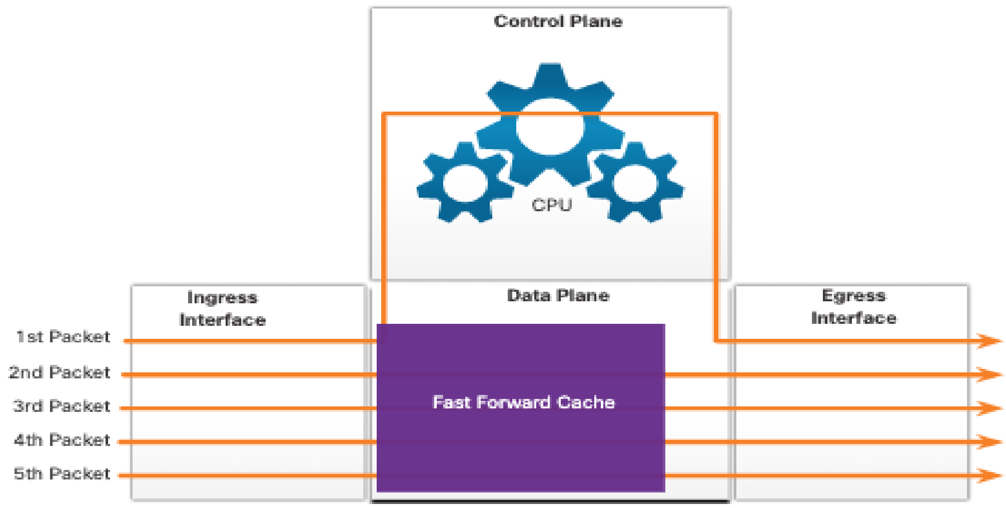
Major Implementation Methods

Different architectures prioritize speed or flexibility. Common mechanisms, notably in Cisco and other hardware routers, include:

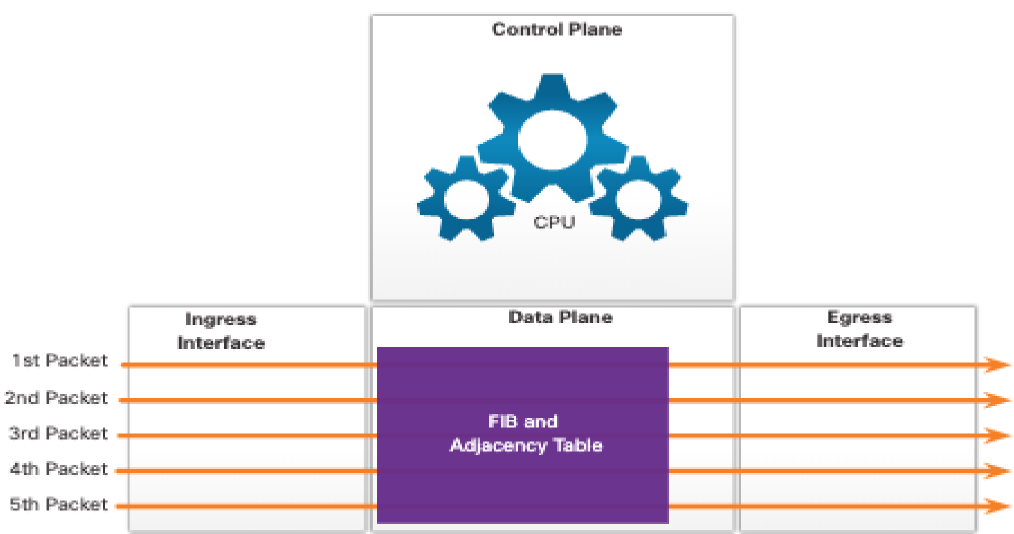
- **Process Switching**: The CPU processes every single packet individually. It is slow because it requires the control plane to handle every packet.



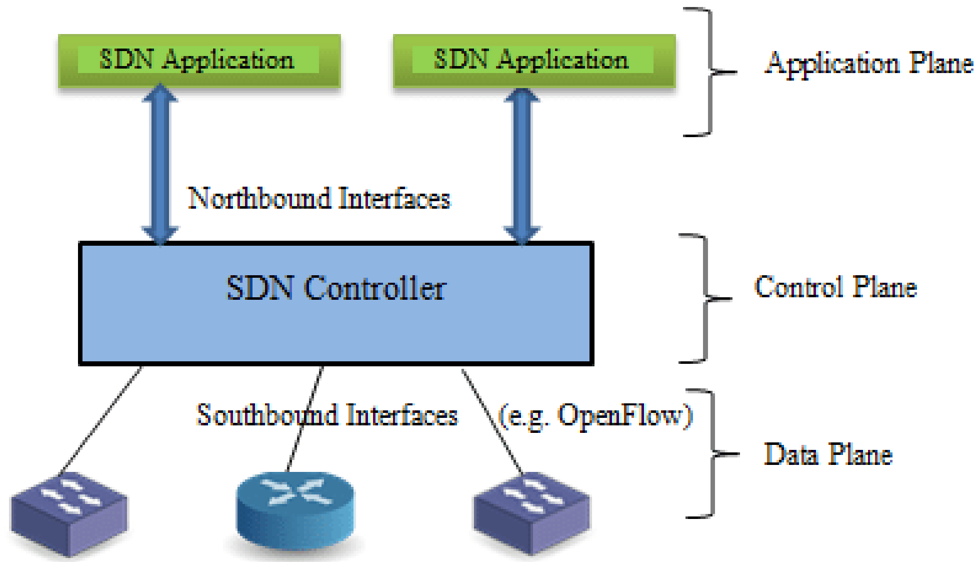
- **Fast Switching**: Only the first packet of a flow is processed by the CPU. The result is cached in a "**fast-switching cache**," allowing subsequent packets in the same flow to be forwarded without CPU intervention.



- **Cisco Express Forwarding (CEF):** The most advanced and default modern method. Its pre-computes all routing information into a **Forwarding Information Base (FIB)** and an **adjacency table** before packets arrive. This allows for line-rate <u>forwarding entirely in hardware</u>.



- **Software-Defined Networking (SDN):** Decouples the **control and data planes**. Forwarding is based on "**flow entries**" pushed to switches by a centralized controller using protocols like **OpenFlow**.



Software defined networking (SDN) is an approach to network management that enables dynamic, programmatically efficient network configuration to improve network performance and monitoring**.** It is a new way of managing computer networks that makes them easier and more flexible to control.

In traditional networks, the hardware (like routers and switches) decides how data moves through the network, but SDN changes this by moving the decision-making to a central software system. This is done by separating the **control plane** (which decides where traffic is sent) from the **data plane** (which moves packets to the selected destination).